Image-Based Measurement Laboratory
Lab3 - Geometry

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Teaching Goals

This third laboratory is concerned with the application of cameras to measurement in a geometric sense. You will do measurements with monocular uncalibrated as well as calibrated sensors.

Important Note!

Please strictly adhere to the following rules when working in the laboratory:

- **Never** use a camera without a suitable mount such as a tripod!
- **Never** touch the unprotected surface of a sensor!

Exercise 1 - Monocular Calibration

An important prerequisite to the use of cameras in vision-based metrology is the metric calibration of the sensor. Calibration refers to the estimation of the model parameters based on one or more acquisitions of a calibration target. We use a pin-hole model which is comprised of the following parameters:

- Radial and tangential distortion.
- Horizontal and vertical focal lengths.
- Skew angle.
- Principal point.

Using a planar calibration target, and a short focal length lens (eg. 6 mm) perform the following points:

- Acquire $N \approx 15$ images of the calibration target under different views.
- Calibrate the camera intrinsics using the Caltech toolbox.
- Visualize the different geometric constellations.
- Discuss your results.
- Visualize the lens distortion by means of a vector field.
- Undistort one of the calibration images and discuss the results.

Exercise 2 - Uncalibrated measurement

In the lab you will find a mobile whiteboard on a stand. You will also be provided with an image of that exact same whiteboard, with some lines drawn on it. Unfortunately those lines have been accidentally erased. Can you help us find out the lengths of the individual lines, as well as the angle between the two angled lines?

Exercise 3 - Calibrated measurement

Using the camera and lens you have calibrated in Exercise 1, take a picture of a ruler using a repro tripod. Also remember the distance the image was taken at (measure to the front of the lens and to the front of the camera body). Now calculate the distance that the image was taken at using the calibration parameters.

Exercise 4 - Fundamental matrix estimation

The fundamental matrix $\boldsymbol{F}$ describes the relationship between two views of the same scene. It is a 3×3 rank 2 matrix and can be estimated using a minimum number of 7 point correspondence. In this exercise you will implement the well-known normalized 8 point algorithm ([2], [3]) to estimate the fundamental matrix. To do so, perform the following two steps:

1. Least squares estimation of $\boldsymbol{F}$: for two corresponding points $\boldsymbol{x_i}$ and $\boldsymbol{x_i'}$ the following equation holds: $\boldsymbol{x_i' F x_i} = 0$. Use this fact to set up a linear equation system of the form $\boldsymbol{Af} = \boldsymbol{0}$ and solve for $\boldsymbol{f}$ which should be the vector representation of $\boldsymbol{F}$.
2. Impose the rank 2 constraint of the fundamental matrix: The most convenient way to impose the rank 2 constraint ($\det \boldsymbol{F} = 0$) is to replace the estimate fundamental matrix $\boldsymbol{F}$ with $\hat{\boldsymbol{F}}$. Let $\boldsymbol{F} = \boldsymbol{USV}^T$ be the singular value decomposition of $\boldsymbol{F}$, with $\boldsymbol{S} = diag(\sigma_1, \sigma_2, \sigma_3)$ being a diagonal matrix with the singular values of $\boldsymbol{F}$. The matrix $\hat{\boldsymbol{F}}$ can be composed according to $\hat{\boldsymbol{F}} = \boldsymbol{U}\hat{\boldsymbol{S}}\boldsymbol{V}^T$, where $\hat{\boldsymbol{S}} = diag(\sigma_1, \sigma_2, 0)$ is the diagonal matrix of the two largest singular values of $\boldsymbol{F}$.

Your task for the exercise is:

- Acquire a stereo image pair of a scene.
- Use 8 corresponding points to estimate $\boldsymbol{F}$.
- Draw epipolar lines to verify the correctness of your estimation.

[1] Multiple View Geometry in Computer Vision. Hartley R. I. and Zisserman A., Cambridge University Press, 2004.

[2] A Computer Algorithm for Reconstructing a Scene from Two Projections. Longuet-Higgins, H. C., 1987

[3] In Defense of the Eight-Point Algorithm. Hartley, Richard I., PAMI, 1997